

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Electrical Engineering Summer Term Examination Academic Year: 2024-25 (Semester – II)	
Class: SY B.Tech	Subject: Signals and Systems	Subject Code: PCC-EE 209
Date: 21.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/sharing of stationery, calculator, etc. is not allowed.
- Non Programmable calculator is allowed.
- Figures to the right indicate full marks.
- The symbols carry the usual meaning.

Q.N.	Attempt The Following Questions	Marks	CO	BT
Q.1	a. Define signal. What are types of standard test Signals.	5	CO1	BT1
	b. Define system and explain basic system properties	5	CO1	BT2
Q.2	a. Find whether given signal is energy or power signal. Also find energy and power of signals. i) $x(t) = e^{j(2\pi t + \frac{3\pi}{4})}$ ii. $x(t) = \cos t$	10	CO1	BT3
	b. For given signal determine whether it is periodic or non periodic. If periodic then find its fundamental period. $x(t) = 4\cos(\frac{\pi t}{100}) + 2\cos(\frac{2\pi t}{180})$	5	CO1	BT3
Q.3	a. Impulse response for discrete time LTI system is given. Check whether system is stable or causal. $h(n) = (\frac{1}{2})^n \cdot u(n)$	7	CO2	BT1
	b. Find the circular convolution of two given sequences $x_1(n)=[1,2,2,1]$, $x_2(n)=[1,2,3,1]$ having period $N=4$.	8	CO2	BT3
Q.4	a. Consider a causal LTI system whose input $x[n]$ and output $y[n]$ are related by difference equation $y(n) = \frac{1}{4} \cdot y(n-1) + x(n)$ Determine $y[n]$ if $x[n] = \delta[n-1]$	8	CO2	BT3
	b. Find the convolution sum of given sequence using matrix method. $x(n) = \{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 0.5, 1 \}$ and $h(n) = \{ \underset{\substack{\uparrow \\ n=0}}{1}, 2, 1, -1 \}$	7	CO3	BT2

Q.5	a.	Determine the Fourier series coefficients. Also find amplitude and phase angle. If $x(t) = 1 + \sin \omega_0 t + 2 \cos \omega_0 t + \cos \left(2\omega_0 t + \frac{\pi}{4} \right)$	7	CO3	BT3
	b.	State properties of continuous time fourier series. Prove any two properties.	8	CO3	BT2
Q.6	a.	Find the fourier transform of following signal. $x(t) = e^{-at} \cdot u(t)$	7	CO3	BT3
	b.	Obtain the step response of the following system represented by impulse response i. $h(t) = t \cdot u(t)$, ii. $h(t) = e^{-3t} \cdot u(t)$	8	CO4	BT2
Q.7	a.	Find the impulse response of the system described by the following difference equation. $y(n) - 3y(n-1) - 4y(n-2) = x(n) + 2x(n-1)$	8	CO4	BT3
	b.	The relation between the input $x(t)$ and output $y(t)$ of causal system is given as $\frac{d^2 y(t)}{dt^2} + \frac{dy(t)}{dt} - 20y(t) = -2x(t) + \frac{5dx(t)}{dt}$ Find impulse response of system.	7	CO4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	To know different types of signals and systems and demonstrate an understanding of characteristics of continuous and discrete -time signals and LTI systems.
CO2	To understand fundamental properties and behaviour of LTI systems and be able to determine response of the system for given input.
CO3	To use the tools (e.g. orthogonal transforms: Fourier transform, Laplace transform, z-transform etc.) for analysis and design of an LTI systems.
CO4	To analyze the behaviour of LTI systems in time and frequency domain using impulse response and transfer function respectively.
CO5	To understand the sampling theorem and the limitations of processing the signals digitally.
CO6	To design a simple LTI system like low-pass or high-pass filters.



Class: SY B.Tech.

Course: Electrical Machines II

Course Code: PCC-EE202/206

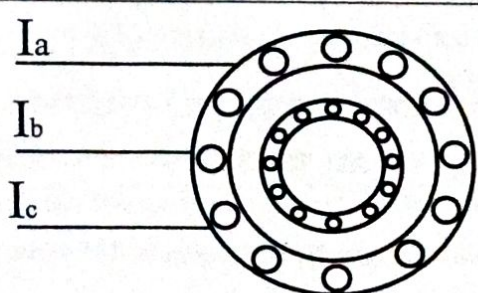
Date: 19/06/2025

Time: 14:30 to 17:30

Max. Marks: 100

Note:

- i. All Questions are compulsory
- ii. Draw waveforms at the end of answer sheets.
- iv. This paper contains 01 pages.

Q.N.		Marks	CO	BT
Q.1	Explain the necessary and essential conditions to produce RMF. Also explain RMF with the help of rotating virtual vector. (Use geometry compass or circle stencil to draw neat diagrams and waveforms)	10	CO1	BT1
Q.2	 Redraw figure very neatly and calculate the total number of slots per pole per phase. Suppose the phase sequence of supply is such that direction of rotation is in clockwise direction. Then assume any one phase, and with the help of stator magnetic line of forces and rotor magnetic line of force, explain the rotation of rotor. (Mark Cross and Dot notations properly for any one phase)	10	CO2	BT2
Q.3	Derive the equation for maximum torque of the induction motor and on the speed torque characteristics show the maximum torque and slip at maximum torque.	10	CO3	BT3
Q.4	With the help of an equivalent electrical model of induction motor, derive all the parameters that are calculated using no load and block rotor test on three phase induction motor.	10	CO3	BT3
Q.5	Explain complete speed torque characteristics of three phase induction motor and explain the power flow for motoring zone, braking zone and generating zone. Assume clockwise rotation in motoring zone.	10	CO3	BT3

Q.6	Explain the induction generator showing power flow of active and reactive power	10	CO3	BT3
Q.7	Explain the electrical model of single-phase induction motor referred to stator side and draw speed torque characteristics of single-phase induction motor.	10	CO3	BT3
Q.8	A 3-phase induction motor draws 50KW from 220V, 50Hz mains. The rotor emf makes 100 oscillations/minutes. If the stator losses are 2kW, how much will the rotor copper loss be? Draw power flow diagram.	10	CO3	BT3
Q.9	A blocked rotor test is performed on three-phase, 40kW, 400V, 50Hz, 6 pole star connected induction motor gave the following data: 200V, 110A, pf=0.4. Determine the starting torque for three-phase voltage of 380V at 45 Hz. Neglect magnetizing current and assume stator and rotor ohmic losses to be equal.	10	CO3	BT3
Q.10	State various starters used to start three phase induction motor. A starting torque of 80 Nm is developed in the induction motor by an auto-transformer starter with a tapping of 30%. If the tapping of the auto-transformers starter is 60%, then what is the starting torque?	10	CO3	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO	On successful completion of this course, students should be able to:
CO 1	Understand the operation of AC machines
CO 2	Implement various testing methods to detect fault.
CO 3	Analyze performance characteristics of ac machines.
CO 4	Implement erection and commissioning of AC Machines as per requirement.
CO 5	Understand the application of special purpose machine.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical Engineering.
Summer Term Examination
Academic Year: 2024-25 (Semester – II)

Class: SY (B.Tech.)	Subject: Electrical Machine - I	Subject Code: PCC-EE 201/202
Date: 24/06/2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
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- v. The symbols carry the usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	Open-circuit and short-circuit tests on a 5 kVA, 220/400 V, 50 Hz, single-phase transformer gave the following results: O.C. test 220 V, 2A, 100 W (l.v. side) S.C. test 40 V, 11.4 A, 200 W (h.v. side) Determine the efficiency and approximate regulation of the transformer at full load 0.9 power factor lagging.	6	CO1	BT3
	b	A d.c. generator has an armature e.m.f. of 100 V when the useful flux per pole is 20 mWb and the speed is 800 r.p.m. Calculate the generated e.m.f. (a) with the same flux and a speed of 1000 r.p.m., (b) with a flux per pole of 24 mWb and a speed of 900 r.p.m.	6	CO1	BT3
	c	Draw the B-H curve and what is the effect on flux density for increasing frequency.	6	CO2	BT4
Q.2	a	Derive for the Referred Values For Primary & Secondary, A 200Kva 1 phase Transformer with Voltage Ratio 6350/660 v As the following Winding Resistance & Reactance $R_1 = 1.56 \Omega$ $R_2 = 0.016 \Omega$ $X_1 = 4.67 \Omega$ $X_2 = 0.048 \Omega$ Referred to high voltage.	7	CO2	BT2
	b	A 16 V, 50 hz ac source is connected to load its draws 30amp current at 0.2lag. When the same load is subjected to 75hz at same voltage what is the new pf of load.	6	CO2	BT3
	c	Draw and explain the critical resistance and critical speed in dc machine	6	CO2	BT3
Q.3	a	A shunt generator delivers 50 kW at 250 V when running at 400 r.p.m. The armature and field resistance are 0.02 ohm and 50 ohm respectively.	6	CO3	BT2

		Calculate the speed of the machine when running as a shunt motor and taking 50 kW input at 250 V. Allow 1 V per brush for contact drop			
	b	A 250 V shunt motor on no load runs at 1000 r.p.m. and takes 5 A. The total armature and shunt field resistance are respectively 0.22 Ω and 250 Ω . Calculate the speed when loaded and taking a current of 50 A, if the armature reaction weakens the field by 3%.	6	CO2	BT3
Q.4	d	A 220 V D.C. Series motor is running at speed of 800 rpm and draws 1000 Amp. Calculate at what speed the motor will run when developing half the torque total resistance of the armature and field is 0.1 ohm Assume that the magnetic ckt is unsaturated. Assume suitable data if necessary.	7	CO3	BT3
	e	Draw and Explain the Characteristics of (shunt or separately) excited and Series dc motor (Brief). Assume suitable data if necessary	6	CO3	BT3
Q.5	a	What are the applications of Brushless DC motor (BLDC)?	6	CO4	BT3
	b	Difference between mechanical & electronic commutator?	6	CO4	BT3
	c	Draw the block diagram of DSP based Brushless direct Current motor drive system.	6	CO4	BT3
Q.2	a	What are the electrical mechanical and economical parameter related to the electric vehicle?	6	CO4	BT2
	b	What are the advantages of brushless DC motor?	6	CO4	BT2
	c	What are the E- motor Configurations in electric vehicle (Multi-motor configurations)?	6	CO4	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

CO 1	Handle and understand static and rotating electrical machines of different types and ratings for installation as per requirement
CO 2	Find out specific rating of AC and DC machines for different application.
CO 3	Develop understanding and applications of various types of electrical machines used in Electrical vehicles.
CO 4	Analyse and evaluate different conditions where the static and rotating electrical machines Can be used.



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Electrical
Summer term Examination
Academic Year: 2024-25 (Semester I)

Class: SY B.Tech.	Subject: Analog Electronics Circuit	Subject Code: PCC-EE-201
Date: 20.06.2025	Time: 14:30 to 17:30	Max. Marks: 100

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.N.		Attempt any 10 questions only	Marks	CO	BT
Q.1	a	Explain P-N junction diode in detail with neat diagrams.	5	CO1	BT2
	b	Explain structure of BJT with its I-V characteristics	5	CO1	BT2
Q.2	a	Explain structure of MOSFET with its I-V characteristics.	5	CO1	BT3
	b	Explain BJT as common emitter amplifier with its I-V characteristics.	5	CO2	BT3
Q.3	a	What is biasing? Explain fixed bias circuit in detail.	5	CO2	BT3
	b	Explain the working of half wave rectifier with neat diagrams.	5	CO2	BT3
Q.4	a	Explain MOSFET as a switch.	5	CO2	BT3
	b	Explain MOSFET as common source amplifier with its I-V characteristics.	5	CO2	BT3
Q.5	a	Differentiate between BJT & MOSFET.	5	CO2	BT2
	b	Explain MOSFET as a common gate amplifier with its I-V characteristics.	5	CO2	BT2
Q.6	a	Explain Class A power amplifier with its efficiency.	5	CO2	BT2
	b	Draw & explain high frequency equivalent circuit of BJT.	5	CO3	BT2
Q.7	a	Draw & explain high frequency equivalent circuit of MOSFET.	5	CO3	BT2
	b	Draw & explain differential amplifier using BJT.	5	CO5	BT3
Q.8	a	Draw & explain block diagram of an op-amp.	5	CO5	BT2
	b	Explain ideal characteristics of op-amp.	5	CO5	BT2
Q.9	a	Define following parameters of op-amp i. Slew rate ii. CMRR iii. Output offset voltage iv. Input offset voltage	5	CO5	BT2

	b	Derive an expression for voltage gain of inverting op-amp with feedback.	5	CO5	BT3
Q.10	a	Explain op-amp as instrumentation amplifier.	5	CO5	BT2
	b	Explain op-amp as integrator.	5	CO5	BT2
Q.11	a	Explain op-amp as zero crossing detector.	5	CO5	BT2
	b	Explain op-amp as square wave generator.	5	CO5	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the current voltage characteristics of semiconductor devices.
CO 2	Develop the ability to analyze and design analog electronic circuits viz. rectifier, amplifier, oscillator, regulator circuits using discrete components.
CO 3	Analyze dc circuits and relate ac models of semiconductor devices with their physical Operation.
CO 4	Evaluate frequency response to understand behavior of electronic circuits.
CO5	Understand the functioning of OP-AMP and design OP-AMP based circuits.
CO6	Take measurement of various analog circuits to compare experimental results in the laboratory with theoretical analysis.